

Quizum: AI-powered Video Summarization and Quiz Generator Website

Specialized Project Report

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- **Context:** Students and self-learners face the challenge of managing and understanding large volumes of video lectures and educational content.
- **Current Approaches:** Manual note-taking, passive video watching, and traditional learning tools provide limited assistance.
- **Limitations:**
 - **Time-Consuming:** Processing lengthy lecture recordings requires significant manual effort.
 - **Inefficient Revision:** 72% of students struggle to revise effectively using long recorded lectures (Online Learning Consortium).
 - **Lack of Interactive Tools:** 67% of instructors believe students need structured summaries and interactive reinforcement tools (EDUCAUSE Review).

Project Goals

To design an intelligent web-based platform (**Quizum**) that transforms educational video lectures into structured learning materials:

Key Objectives

- 1 **Automatic Transcription:** Use Whisper AI to convert speech to text with timestamp segments.
- 2 **AI-Powered Summarization:** Generate concise summaries using fine-tuned PEGASUS model.
- 3 **Quiz Generation:** Create self-assessment quizzes using Flan-T5 for interactive learning.
- 4 **Improve Learning Efficiency:** Enhance study efficiency, learner engagement, and knowledge retention.

VideoKen - Interactive video learning platform for enterprise L&D
Key Features:

- **Auto-generated chapters:** AI analyzes visual/audio to segment videos
- **Phrase clouds:** Navigate to specific topics instantly
- **Deep search:** Search spoken words across video library
- **In-video quizzes:** Manual embedding of assessments

Limitations:

- Focus on navigational indexing
- Manual quiz creation (not automated)
- Enterprise-focused, not education
- No automated assessment generation

Otter.ai - Meeting transcription

- Real-time transcription
- Speaker identification
- Executive summaries
- Keyword extraction
- **Limitation:** No quiz generation

Kwizie - Gamified quiz platform

- AI quiz generation from videos
- Multiplayer game format
- Microlearning approach
- LLM-powered MCQ creation
- **Limitation:** No summarization

Quizum's Advantage: Integrates transcription + summarization + quiz generation in one cohesive workflow

Related Work: Comparison

Feature	VideoKen	Otter.ai	Kwizie
Speech-to-Text	Yes	Yes	Yes
Auto Summarization	Phrase clouds	Yes	No
Auto Quiz Gen	No	No	Yes
Interactive Quizzes	Yes	No	Yes
AI Chatbot/Q&A	Search-based	Yes	No
Target Audience	Enterprise	Business	Education

Table: Comparison of related platforms

Quizum uniquely combines: Transcription + Summarization + Quiz Generation + Interactive Learning

Stakeholders & Their Needs

- **Students (Primary Users):**

- Time-saving platform for quick revision
- Accurate, concise summaries
- Accessibility across devices

- **Educators:**

- Insights into student engagement
- Identify knowledge gaps
- Ensure information accuracy

- **Developers:**

- Scalable, performant system
- Clear requirements and feedback loops
- Seamless integration capabilities

- **Content Creators:**

- Intellectual property protection
- Increased lecture impact and accessibility

System Benefits

Key Benefits for All Stakeholders: For Learners:

- **Time savings:** Quick revision without rewatching
- **Better retention:** 25% improvement with summaries + quizzes
- **Personalized learning:** Focus on weak areas
- **Interactive engagement:** Active vs passive learning

Institutional Value: Improved student performance, innovation in digital learning, scalable platform

For Educators:

- Track student interaction patterns
- Identify knowledge gaps
- Enhanced teaching effectiveness
- Reduced manual quiz creation

① Video Input & Preprocessing:

- Accept URL (YouTube) or file upload (MP4, AVI, MOV; max 500MB)
- Transcribe with Whisper (85%+ accuracy on clear audio)
- Preserve timestamps for segment linking

② Summarization Module:

- Generate extractive/abstractive summaries (10-20% of original)
- Extract main topics, key points, conclusions
- Output: Plain text, bullets, or PDF with timestamps

③ Quiz Generation:

- Auto-generate 5-15 questions per video
- Types: MCQ (4 options), true/false, fill-in-blank
- Include correct answer + explanation

Non-Functional Requirements

Performance:

- Process 10-min video in under 2 minutes (transcription + summarization + quiz)
- Handle videos up to 2 hours without crashing
- **Accuracy targets:**
 - Transcription: 85%+ (WER metric)
 - Summary coherence: 80%+ (ROUGE score)
 - Quiz relevance: 90%+ (human-evaluated)

Usability:

- No technical expertise required
- Tooltips, tutorials, demo videos
- Default English, with localization options

Security & Reliability:

- User data protection and privacy
- 99% uptime target

UC-01: Upload Video

- User uploads video/link
- System validates format
- Queues for transcription

UC-02: Summarize Video

- Retrieves transcript
- Generates summary
- Displays to user

UC-03: Generate Quiz

- Analyzes transcript
- Creates questions
- Stores for practice

UC-04: Take Quiz

- User answers questions
- System evaluates
- Provides feedback

UC-05: Download/Export Results

- User selects format
- System generates file
- User downloads content

The system implements a modern, decoupled client-server architecture with three primary layers:

① Frontend Layer:

- Next.js (React-based)
- Server-side rendering (SSR)
- Real-time video playback

② Backend Layer:

- Node.js + Express
- RESTful API
- Request validation

③ Worker Layer:

- BullMQ + Redis queue
- Async AI processing

System Architecture Diagram

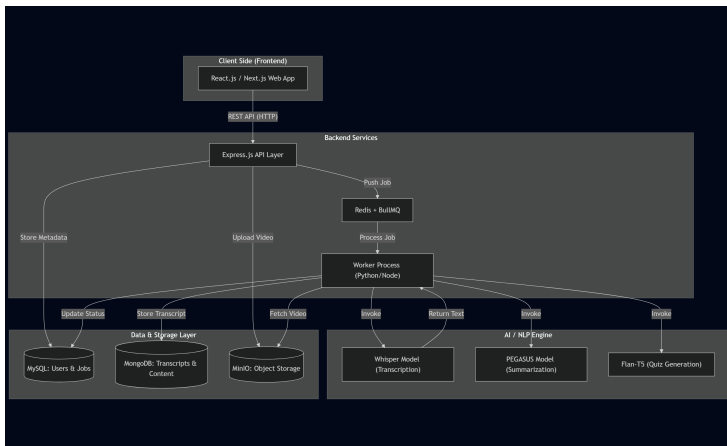


Figure: System Architecture Diagram

The worker service orchestrates three AI models for content processing:

① **Whisper (Speech-to-Text):**

- Robust transcription with timestamp segments
- Multiple model sizes (Tiny to Large)
- Selected: Whisper Small (optimal balance)

② **PEGASUS (Summarization):**

- Fine-tuned on SAMSum dataset (14,732 conversations)
- Abstractive summarization for dialogue
- Generates concise, coherent summaries

③ **Flan-T5 (Quiz Generation):**

- Instruction-tuned model
- Generates multiple-choice questions
- Links questions to transcript segments

Data Storage Strategy

The system uses a **polyglot persistence** approach to handle different data types efficiently:

- **MySQL (Relational):**
 - Job tracking and status management
 - User accounts and authentication
 - Quiz metadata and attempts
 - Structured queries and consistency
- **MongoDB (Document):**
 - Full transcripts with segments
 - Flexible schema for varying content
 - Efficient for large text documents
- **MinIO (Object Storage):**
 - S3-compatible storage
 - Video and audio assets
 - Staging for future AWS S3 migration

Database Schema: MySQL

Core Tables:

- **jobs:** UUID, status (ENUM), payload (JSON), result (JSON)
- **videos:** id, owner_id, storage_key, status
- **users:** email, password_hash, role

Assessment Tables:

- **quizzes:** video_id (FK), type
- **questions:** text, options (JSON), answer, source_segments
- **quiz_attempts:** score, details (JSON)

Key Features:

- Foreign key relationships for data integrity
- JSON columns for flexible data (payload, options, details)
- Status tracking with ENUM types for consistency
- Source_segments link questions to video timestamps

MySQL Database Schema

images/mysql_schema.png

Transcript Collection:

- **videoid:** Link to MySQL job
- **fullText:** Complete transcription
- **segments:** Array of {start, end, text}

Why MongoDB?

- Flexible schema for varying content
- Efficient for large text documents
- Supports full-text search

Data Flow:

- 1 Upload → Redis queue (volatile)
- 2 Worker processes job
- 3 Save transcript to MongoDB
- 4 Update job status in MySQL

Hybrid Strategy Benefits:

- Fast structured queries (MySQL)
- Efficient content retrieval (MongoDB)
- Optimized for both use cases

Whisper Model Evaluation

Setup:

- **Test Audio:** 30-minute Yale lecture (PSYC 110)
- **Hardware:** Tesla T4 GPU (16GB), 12.7GB RAM
- **Platform:** Google Colab
- **Models Tested:**
 - Tiny (190MB)
 - Base (285MB)
 - Small (1094MB)
 - Medium (3309MB)
 - Large (4649MB)

Metrics:

- **WER:** Word Error Rate
- **CER:** Character Error Rate
- **Speed:** Inference time (seconds)
- **Memory:** Model size

Results: Accuracy vs Speed Trade-off

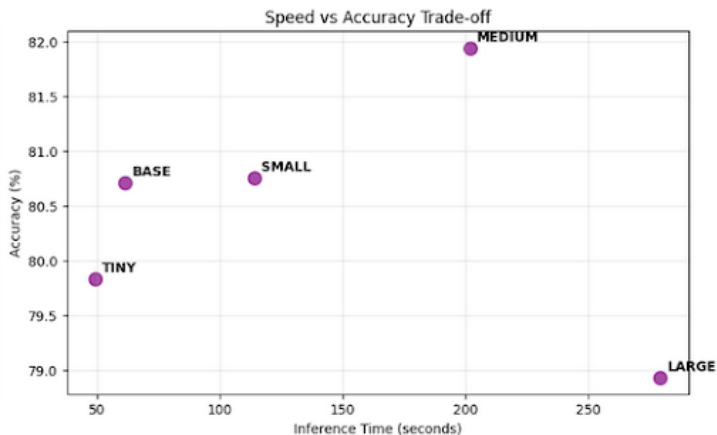
Whisper Small provides the best balance of accuracy, speed, and memory usage.

Model	WER (%)	CER (%)	Speed (s)
Tiny	20.02	16.90	49.5
Base	19.22	16.50	61.6
Small	18.68	16.47	114.1
Medium	18.27	16.36	201.7
Large	19.58	17.69	279.2

Table: Whisper Model Performance Comparison

Key Finding: Whisper Small selected as optimal model - only 0.4% WER difference from Medium, but **43% faster** (114s vs 201s).

Results: Accuracy vs Speed Trade-off (Chart)



PEGASUS Summarization Results

Training Details:

- **Dataset:** SAMSum (14,732 conversations)
- **Model:** Google PEGASUS
- **Task:** Dialogue summarization
- **Approach:** Fine-tuning for conversational text

Performance:

- Generates concise summaries
- Maintains coherence
- Captures key dialogue points
- Optimized for lecture transcripts

PEGASUS Summarization Performance

```
Device set to use cuda
Your max_length is set to 128, but your Input_length is only 122. Since this is a summarization task, where outputs shorter than the input are typically wanted
Dialogue:
Hannah: Hey, do you have Betty's number?
Amanda: Lemme check.
Hannah: <file_gif>
Amanda: Sorry, can't find it.
Amanda: Ask Larry
Amanda: He called her last time we were at the park together
Hannah: I don't know him well
Hannah: <file_gif>
Amanda: Don't be shy, he's very nice
Hannah: If you say so..
Hannah: I'd rather you texted him
Amanda: Just text him 🍷
Hannah: Orgh.. Alright
Hannah: Bye
Amanda: Bye bye

Reference Summary:
Hannah needs Betty's number but Amanda doesn't have it. She needs to contact Larry.

Model Summary:
Larry called Betty's number last time they were at the park together. He's very nice. Hannah wants Amanda to text him, but she can't find Betty's number.
```

Figure: PEGASUS training and validation performance

Flan-T5 Quiz Generation Experiments

Experimental Setup:

- **Model:** google/flan-t5-base
- **Method:** PEFT (LoRA) for efficient fine-tuning
- **Dataset:** RACE (Reading Comprehension from Examinations)
- **Goal:** Generate valid MCQ with Article, Question, Options, and Answer

Zero-Shot Baseline:

- Initial performance was poor
- Generated malformed, repetitive outputs
- Failed to follow MCQ structure

```
Which of the following is NOT true?Options:A Plants convert light energy into chemical energy through photosynthesis.  
B) Plants convert light energy into chemical energy through photosynthesis.  
C) Plants convert light energy into chemical energy through photosynthesis.  
D) Plants convert light energy into chemical energy through photosynthesis.  
Answer: A
```

Flan-T5 Fine-Tuning Results

Training Outcome (1 Epoch, $\approx 62k$ examples):

- **Syntactic Success:** Model successfully learned the MCQ format (generating distinct options A, B, C, D).
- **Semantic Challenge:** Still struggles with deep reasoning to consistently identify the *correct* answer.

```
--- Starting REAL Training Run (1 Epoch on FULL data) ---  
[15612/15612 3:00:22, Epoch 1/1]  
Epoch Training Loss Validation Loss  
1 0.420000 0.379522  
--- REAL Model Saved! Model saved to /content/gdrive/MyDrive/models/flan-t5-mcq-adaptor ---
```

Figure: Flan-T5 generation examples after fine-tuning

Backend Test: Video Upload API

Testing POST /api/upload

Upload Audio/Video



KET2-2.MP3

Supported: audio and video files

Size: 1.41 MB

Upload and transcribe

Validates file upload handling, MinIO storage integration, and initial job creation in MySQL.

Backend Test: Job Management

Job Listing (GET /api/jobs)

Transcription Jobs

KET2-2.MP3

Job ID: 20

Video ID: 20

Created: 12/1/2025, 5:48:38 PM

completed 

Linear Regression in 3 Minutes.mp4

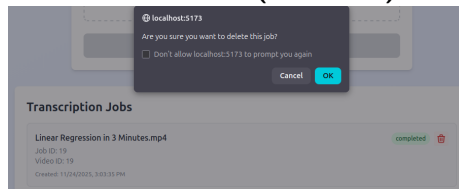
Job ID: 19

Video ID: 19

Created: 11/24/2025, 3:03:35 PM

completed 

Job Deletion (DELETE)



Verifies pagination metadata and cascading deletion of job records and transcripts.

Backend Test: Transcript Retrieval

Testing GET /api/transcripts/:videoid

Transcript

Linear Regression in 3 Minutes.mp4



AI Summary

Generate Summary

Full Transcript

Linear regression is a statistical technique for modeling the relationship between an output variable and one or more input variables. In layman's terms, think of it as fitting a line through some data points as shown here, so you can make predictions on unknown data, assuming there is a linear relationship between the variables. You might be familiar with the linear function y equals mx plus b , where y is the output variable, also called the dependent variable. You may also see expressed as f of x , the function of the input variable. x on the other hand, would serve as the input variable, also called the independent variable. And slightly you'll see the coefficients m and b expressed as β_1 and β_0 respectively. So what do the m and b coefficients do? The m or β_1 coefficient controls the slope of the line, the b or the β_0 controls the intercept of the line, in machine learning we also know it as the bias. These two coefficients are what we are solving for in linear regression. We can also extend to multiple input variables, so x_1, x_2, x_3 , with β_1, β_2 , and β_3 , and so on, acting as slopes for each of those variables. In these higher dimensions, you would visualize the linear regression as a hyperplane. So how do we fit the line to these points? Well you'll notice that there's these differences between the points and the line, these little red segments, these are called residuals. They are the differences between the data points and the predictions the line would produce. Take each of these residuals and square them. These are the squared errors, and notice that the large of the residuals are, the more amplified area of the squares are. If we total the areas of all of these squares for a given line, we will get the sum of the squared error, and this is known as our loss function. We need to find the β_0 and β_1 coefficients that will minimize that sum of squared error. The coefficients can be solved with a variety of techniques ranging from matrix decomposition to gradient descent, which is depicted right here. Thankfully a lot of libraries are available to do this for us, and we will deep dive into these topics in other videos. To validate a linear regression, there are a number of techniques. Machine learning practitioners will often take a third of the data and put it into the test data set. The remaining two thirds will become the training data set. The training data set will then be used to fit the regression line. The test data set will then be used to validate the regression line. This is done to make sure that the regression performs well on data it has not seen before. The tricks used to evaluate the linear regression vary from the R square, standard error of the estimate, prediction intervals, as well as statistical significance. These are topics we will cover in future videos. If you enjoyed this video, please like and subscribe. Look at my two O'Reilly books, Essential Math for Data Science, and getting started with SQL. Chapter five of Essential Math for Data Science actually covers linear regression and much more depth. If you want live instruction, I also do teach on the O'Reilly platform. Promotional link below. I teach classes including machine learning from scratch, probability, and SQL. Comment on what topics you would like to see next, and I will see you again on 3 Minute Data Science.

Segments



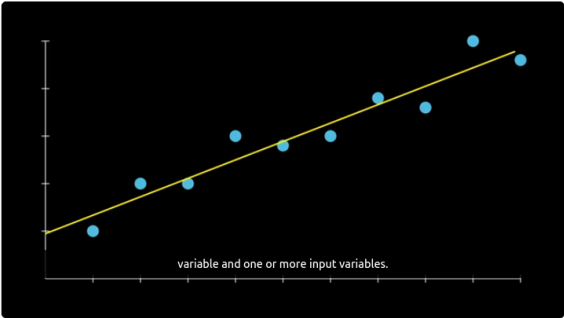
Backend Test: Job Details & Playback

Testing GET /api/jobs/:jobId

Transcript

Linear Regression in 3 Minutes.mp4

× Hide Video Player



variable and one or more input variables.

Transcript Segments (Click to jump)

Time Range	Text
0:00 - 0:20	Lenny regression is a statistical technique for modeling the relationship between an output
0:20 - 0:23	variable and one or more input variables.

Backend Test: Transcript Retrieval

Testing GET /api/transcripts/:videoid

Transcript

Linear Regression in 3 Minutes.mp4



AI Summary

Generate Summary

Full Transcript

Linear regression is a statistical technique for modeling the relationship between an output variable and one or more input variables. In layman's terms, think of it as fitting a line through some data points as shown here, so you can make predictions on unknown data, assuming there is a linear relationship between the variables. You might be familiar with the linear function y equals mx plus b , where y is the output variable, also called the dependent variable. You may also see expressed as f of x , the function of the input variable. x on the other hand, would serve as the input variable, also called the independent variable. And slightly you'll see the coefficients m and b expressed as β_1 and β_0 respectively. So what do the m and b coefficients do? The m or β_1 coefficient controls the slope of the line, the b or the β_0 controls the intercept of the line, in machine learning we also know it as the bias. These two coefficients are what we are solving for in linear regression. We can also extend to multiple input variables, so x_1, x_2, x_3 , with β_1, β_2 , and β_3 , and so on, acting as slopes for each of those variables. In these higher dimensions, you would visualize the linear regression as a hyperplane. So how do we fit the line to these points? Well you'll notice that there's these differences between the points and the line, these little red segments, these are called residuals. They are the differences between the data points and the predictions the line would produce. Take each of these residuals and square them. These are the squared errors, and notice that the large of the residuals are, the more amplified area of the squares are. If we total the areas of all of these squares for a given line, we will get the sum of the squared error, and this is known as our loss function. We need to find the β_0 and β_1 coefficients that will minimize that sum of squared error. The coefficients can be solved with a variety of techniques ranging from matrix decomposition to gradient descent, which is depicted right here. Thankfully a lot of libraries are available to do this for us, and we will deep dive into these topics in other videos. To validate a linear regression, there are a number of techniques. Machine learning practitioners will often take a third of the data and put it into the test data set. The remaining two thirds will become the training data set. The training data set will then be used to fit the regression line. The test data set will then be used to validate the regression line. This is done to make sure that the regression performs well on data it has not seen before. The tricks used to evaluate the linear regression vary from the R square, standard error of the estimate, prediction intervals, as well as statistical significance. These are topics we will cover in future videos. If you enjoyed this video, please like and subscribe. Look at my two O'Reilly books, Essential Math for Data Science, and getting started with SQL. Chapter five of Essential Math for Data Science actually covers linear regression and much more depth. If you want live instruction, I also do teach on the O'Reilly platform. Promotional link below. I teach classes including machine learning from scratch, probability, and SQL. Comment on what topics you would like to see next, and I will see you again on 3 Minute Data Science.

Segments



Testing POST /api/transcripts/:videoid/summarize

AI Summary

Regenerate

Lenny regression is a statistical technique for modeling the relationship between an output variable and one or more input variables. <n> To validate a linear regression, machine learning practitioners will often take a third of the data and put it into the test data set. The remaining two thirds will become the training data set. The test data set will then be used to validate the regression line.

Triggers the PEGASUS model to generate an abstractive summary of the transcript.

- ① **Contribution:** Successfully implemented an end-to-end AI-powered learning platform.
- ② **Technical Achievements:**
 - Integrated Whisper (Small) for optimal transcription
 - Fine-tuned PEGASUS on SAMSum for dialogue summarization
 - Implemented polyglot persistence (MySQL + MongoDB + MinIO)
 - Built async processing pipeline with BullMQ + Redis
- ③ **Impact:**
 - Transforms passive video watching into active learning
 - Reduces manual effort in study preparation
 - Provides interactive quizzes for self-assessment

Planned Enhancements

- **MCQ Generator model:** Upgrade to a larger model for multiple difficulty levels and high diversity.
- **AWS S3 Migration:** Transition from MinIO to AWS S3 for improved reliability and scalability.
- **Cloud Deployment:** Deploy to production environment with CI/CD pipeline.
- **New Features:**
 - Automatic mind-map generation from transcripts
 - Integrated AI chatbot for Q&A and content guidance
 - Flashcard generation for reinforcement learning

Q & A

Thank you for your attention!